

Marketing Funnel & Conversion Performance Analysis

1. Project Overview

This project focused on analyzing a bank marketing campaign dataset to understand customer conversion behavior and marketing funnel performance.

The goal of the analysis was to evaluate how customers moved through the marketing funnel, identify conversion patterns, determine where drop-offs occurred, and provide actionable recommendations to improve campaign performance.

The project simulated real-world marketing analytics work commonly performed in:

- Banking institutions
- Fintech companies
- Marketing agencies
- SaaS companies
- Growth analytics teams

The analysis was designed to answer important business questions such as:

- Which customer groups convert the most?
- Which campaign strategies perform best?
- How does contact frequency impact conversion?
- Which customer segments should be targeted more effectively?
- Where are customers dropping off in the funnel?

2. Business Problem

Banks invest heavily in marketing campaigns to attract customers and encourage them to subscribe to financial products such as term deposits.

However, not all customers respond positively to campaigns.

The business challenge was to:

- Improve customer conversion rates
- Reduce campaign inefficiencies

- Understand customer behavior
- Identify high-performing customer segments
- Improve future marketing strategies using data-driven insights

This analysis aimed to transform raw campaign data into actionable business intelligence.

3. Dataset Information

Dataset Used

Bank Marketing Campaign Dataset

Dataset Source

- UCI Machine Learning Repository
- Kaggle

Dataset Description

The dataset contains customer demographic information, marketing campaign interactions, previous campaign outcomes, and customer conversion status.

The dataset was used to analyze:

- Customer conversion behavior
- Funnel performance
- Campaign effectiveness
- Customer segmentation
- Marketing optimization opportunities

4. Data Cleaning & Preparation

The dataset was initially imported as a CSV file where all values appeared in a single column due to semicolon delimiters.

The following data cleaning and preparation steps were performed in Microsoft Excel:

Cleaning Steps Performed

- Split CSV data correctly using semicolon delimiters
- Converted the dataset into a structured Excel table

- Checked for missing/null values
- Verified and corrected data types
- Checked for duplicate rows across all columns
- Removed exact duplicate records
- Checked categorical text consistency
- Preserved valid “unknown” values for business analysis

Helper Columns Created

Additional helper columns were created to improve dashboard analysis and customer segmentation.

Age Group

Grouped customers into:

- Under 30
- 30–40
- 41–50
- 51+

Conversion Status

Converted the target variable into:

- Converted
- Not Converted

Contact Frequency

Grouped customers based on campaign contact attempts:

- Single Contact
- 2–3 Contacts
- 4+ Contacts

Campaign Outcome Group

Grouped previous campaign outcomes into:

- Successful Campaign
- Failed Campaign

- No Previous Outcome

5. Power BI Development Process

The cleaned dataset was imported into Microsoft Power BI for analysis and dashboard development.

Data Modeling

The dataset was structured for analytical reporting and KPI generation.

DAX Measures Created

Several DAX measures were developed to calculate key marketing metrics.

Total Customers

Calculated the total number of customers contacted during campaigns.

Total Converted

Calculated the total number of customers who successfully converted.

Total Not Converted

Calculated customers who did not convert.

Conversion Rate

Calculated the percentage of customers who converted successfully.

6. Dashboard Development

A multi-page interactive Power BI dashboard was developed to visualize funnel performance and customer conversion behavior.

PAGE 1 – Funnel Overview

Objective

Provide a high-level overview of campaign performance and marketing funnel effectiveness.

Visuals Included

KPI Cards

- Total Customers
- Total Converted
- Conversion Rate
- Total Not Converted

Funnel Chart

Displayed the overall customer funnel:

- Contacted Customers
- Converted Customers

Donut Chart

Compared:

- Converted customers
- Not converted customers

Conversion Rate by Month

Analyzed conversion trends across different months.

Customer Conversion by Contact Frequency

Analyzed how the number of campaign contacts affected customer conversion.

Interactive Slicers

Users could filter the dashboard using:

- Month
- Job
- Age Group

PAGE 2 – Customer Segment Analysis

Objective

Analyze customer demographics and identify high-performing customer segments.

Visuals Included

Conversion by Age Group

Analyzed which age groups converted the most.

Conversion by Job Role

Compared customer conversions across professions.

Conversion by Marital Status

Analyzed conversion performance by relationship status.

Conversion by Education Level

Evaluated how education levels impacted conversion.

Conversion by Campaign Outcome Group

Analyzed how previous campaign success affected future conversion behavior.

7. Key Insights & Findings

The analysis revealed several patterns in customer conversion behaviour and campaign performance.

Customer Drop-Offs in the Funnel

The funnel analysis showed that a large number of customers were contacted during the campaign, but only a smaller portion converted successfully. This indicates that there are significant drop-off points between customer engagement and final conversion.

The results suggest that campaign targeting and customer engagement strategies could be improved to increase conversion performance.

Monthly Conversion Performance

The months with the highest conversion levels were:

- March
- September
- October
- December

These periods showed stronger campaign performance than other months, which may indicate seasonal customer behaviour or more effective campaign timing.

Customer Segment Insights

Job Role

Customers working in administrative roles converted more frequently than technicians and several other professions.

Age Group

Customers aged 30–40 recorded the highest conversion levels, while customers aged 51+ showed lower conversion activity.

Marital Status

Married customers showed the highest conversion levels, followed by single customers.

Education Level

Customers with university degrees converted more often than customers with only high school education.

These findings suggest that customer demographics play an important role in campaign performance and should be considered when planning future marketing strategies.

Contact Frequency Analysis

The dashboard showed that many customers converted after only one contact attempt.

Customers who were contacted multiple times generally showed lower conversion performance. This may suggest that excessive contact attempts reduce customer engagement.

Previous Campaign Outcome Analysis

The campaign outcome analysis focused on previous customer interactions with marketing campaigns.

Customers grouped under “No Previous Outcome” recorded the highest conversion levels. This means that many customers who converted during the campaign had not participated in previous campaigns before.

This finding suggests that targeting new or previously uncontacted customers may create strong conversion opportunities.

Overall Findings

The project showed that conversion performance is influenced by several factors, including:

- customer demographics
- campaign timing
- contact frequency
- previous campaign history

The dashboard helped transform raw marketing campaign data into clear business insights that can support future campaign optimization and data-driven decision-making.

8. Business Recommendations

Based on the findings, several recommendations were proposed to improve future campaign performance.

Recommendation 1 – Reduce Excessive Contact Frequency

Over-contacting customers may reduce engagement and conversion rates.

The business should optimize campaign contact frequency to avoid customer fatigue.

Recommendation 2 – Focus on High-Converting Customer Segments

The bank should prioritize marketing efforts toward:

- High-performing age groups
- Strong-converting professions
- Customers with positive previous campaign history

Recommendation 3 – Re-Target Previously Successful Customers

Customers who responded positively to previous campaigns are more likely to convert again. These customers should be prioritized for future campaigns.

Recommendation 4 – Improve Low-Performing Campaign Periods

Campaigns during weaker-performing months should be reviewed and optimized.

Recommendation 5 – Use Customer Segmentation for Personalized Marketing

Customer demographics and behavior should be used to create more targeted and personalized campaigns.

9. Skills Demonstrated

This project helped demonstrate and strengthen the following technical and analytical skills:

Technical Skills

- Microsoft Power BI
- Power Query
- DAX Measures
- Data Cleaning
- Data Transformation
- Dashboard Development
- Data Visualization
- Excel Data Preparation

Analytical Skills

- Funnel Analysis
- Marketing Analytics
- KPI Development

- Conversion Analysis
- Customer Segmentation
- Business Insight Generation
- Data-Driven Decision Making

Business Skills

- Executive Reporting
- Storytelling with Data
- Marketing Performance Analysis
- Strategic Recommendation Development

10. How To Present This Project to Recruiters

Short Recruiter Introduction

“I completed a Marketing Funnel & Conversion Performance Analysis project as part of my Data Science & Analytics Internship with Future Interns.

The project focused on analyzing a bank marketing campaign dataset to understand customer conversion behavior, campaign effectiveness, and funnel performance.

Using Power BI, Power Query, Excel, and DAX, I built an interactive dashboard that tracked conversion metrics, identified drop-off points, analyzed customer segments, and generated business recommendations to improve campaign performance.

The analysis helped identify which customer groups converted the most, how campaign contact frequency impacted conversion, and how previous campaign outcomes influenced future customer behavior.”

11. What Makes This Project Valuable

This project demonstrates the ability to:

- Work with real-world business datasets
- Clean and transform raw data
- Build professional dashboards
- Analyze customer behavior
- Generate actionable business insights

- Present findings visually to stakeholders

The project combines technical analytics skills with business problem-solving and marketing strategy analysis.